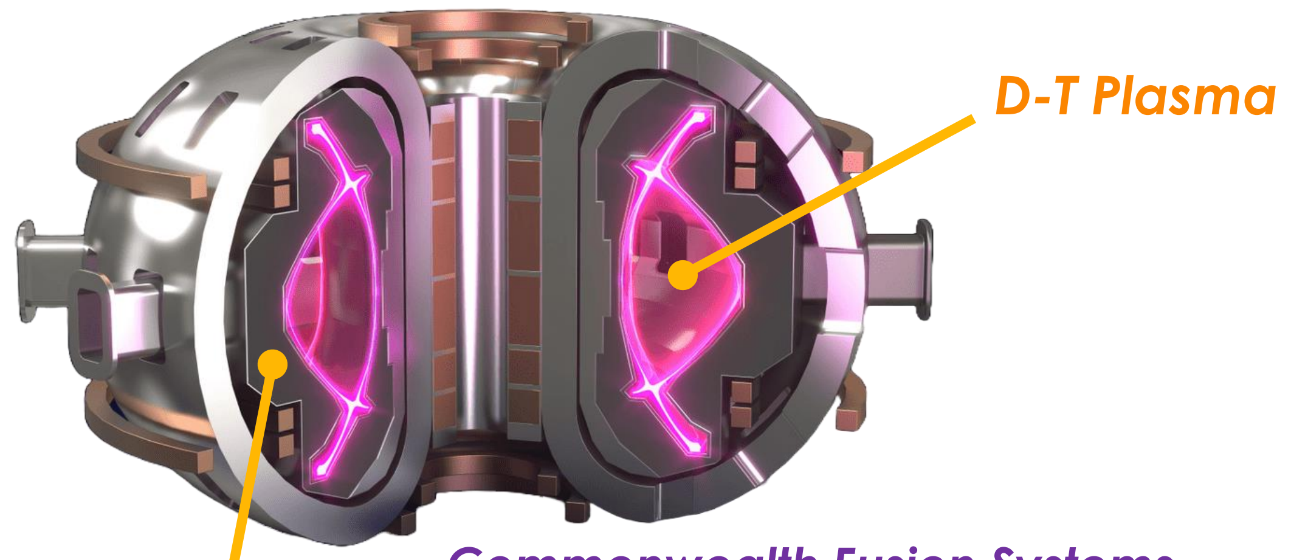
FUELING THE FUSION FUTURE

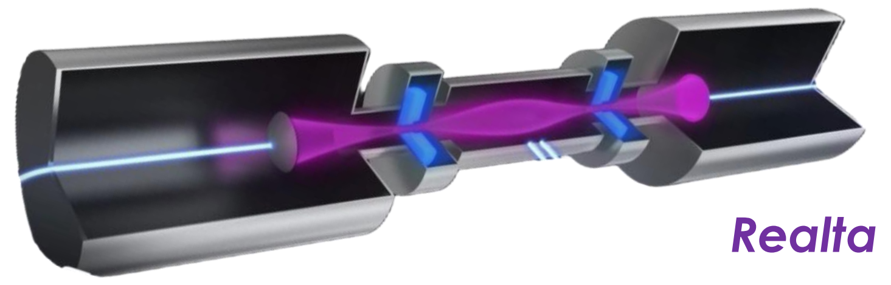
Tritium and deuterium are likely going to fuel the first fusion power plants...

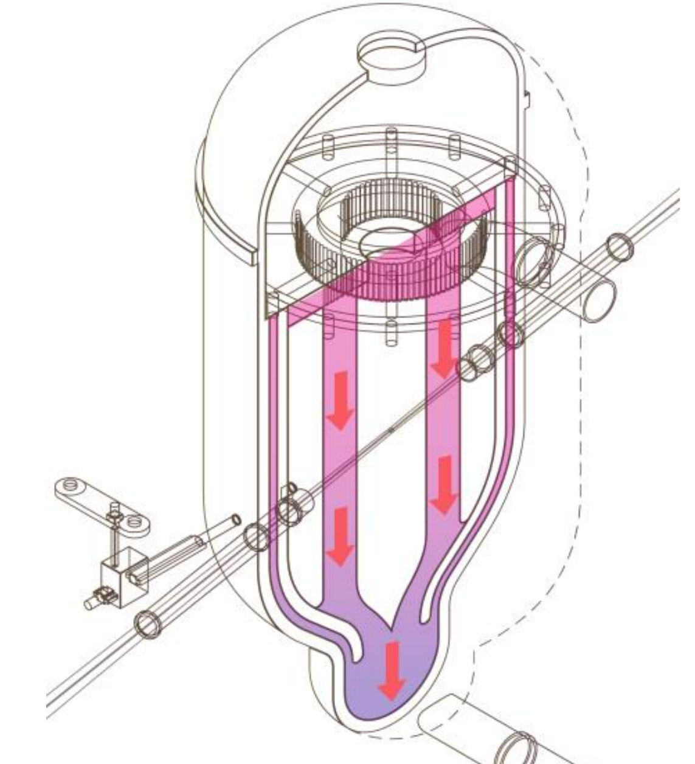
$$D + T \rightarrow He + n \qquad \begin{array}{l} \boxed{n} + {}^{6}Li \rightarrow \boxed{T} + He + 4.8\ \text{MeV} \\ \boxed{n} + {}^{7}Li \rightarrow \boxed{T} + He + n - 2.5\ \text{MeV} \end{array}$$

...but while deuterium is abundant, tritium needs to be constantly regenerated.

The **Liquid Immersion Blanket** (LIB) is a proposed tank of molten salt surrounding the plasma core that provides a combined solution for tritium breeding, heat extraction, and neutron shielding in a fusion device. The LIB technology can be adapted to numerous reactor concepts.







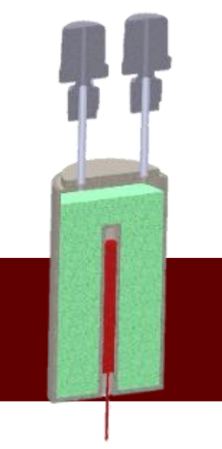
The Liquid Immersion Blanket Robust Accountancy project aims to comprehensively de-risk the Liquid Immersion Blanket technology.

We are irradiating **FLiBe** and **ClLiF molten salts** with **14 MeV fusion-spectrum neutrons** from a sealed-tube source. To gain operational experience, build subsystems, neutron and tritium sensing techniques, as well as to optimize experimental design, we are employing a **staged approach**. Currently, we have more than **two years of experience breeding tritium**, becoming the first group to **experimentally measure the Tritium Breeding Ratio**.

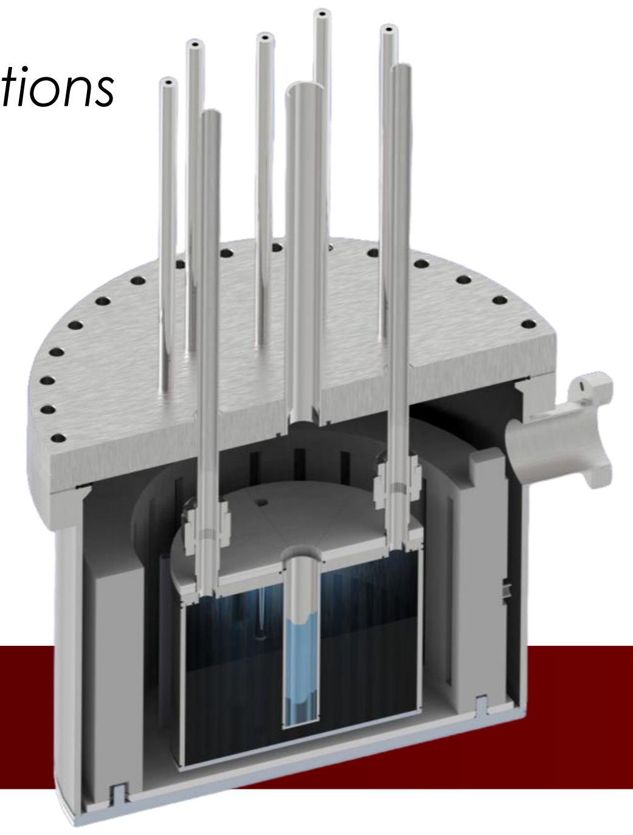
Benchtop scale experiments *enable rapid iterative testing of:*

- *Tritium transport and speciation*
- *Salt electrochemistry*
- *Multiple breeder compositions*

These results help in model benchmarking and inform large-scale design.



BABY - 0.1 L
completed



BABY - 1 L
operating

Medium-scale testing *helps us gain experience in:*

- *Bulk salt handling*
- *Managing permeation*
- *Navigating structural and thermal constraints*

*The **Pi** tank is designed to cover π steradians around the point neutron source… while also resembling a slice of **pie**.*

LIBRA Pi - 100 L
fabricated

A net-tritium breeding experiment *is the aspirational goal of LIBRA. Our largest test is designed to achieve a **Tritium Breeding Ratio** greater than one – a necessary prerequisite for any self-sustaining fusion device.*

$$\mathbf{TBR} = \frac{\text{tritons created}}{\text{tritons consumed}}$$

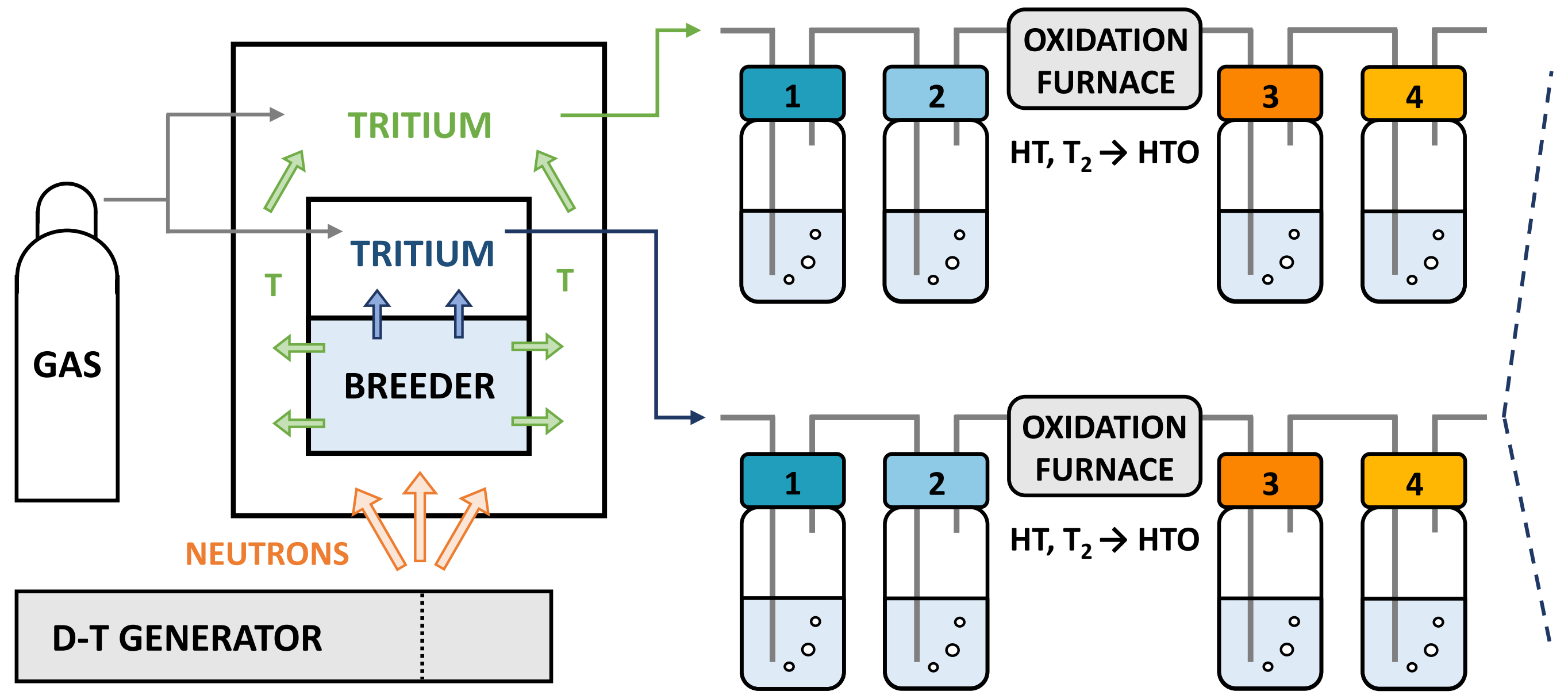
*A **confluence of hazards**, both radiological (neutrons, tritium) and chemical (beryllium, HF), made it necessary to build a dedicated facility and personnel experience in-house. This gave rise to the nearly completed **MIT Molten Salt Laboratory**.*

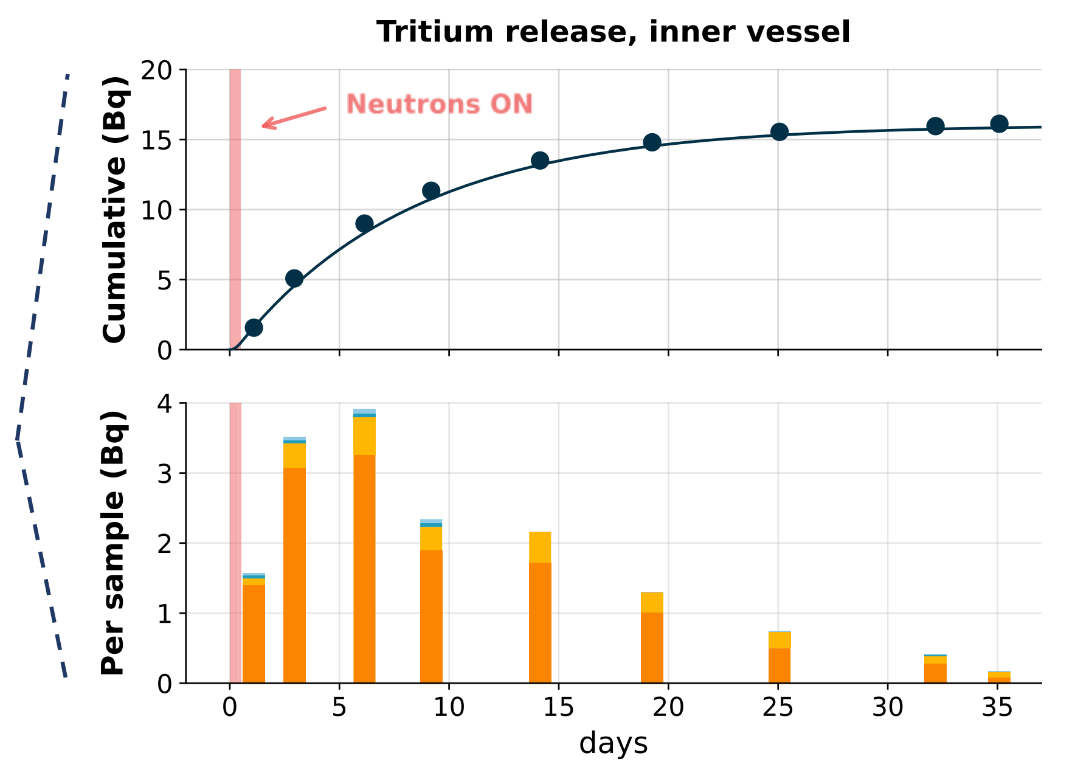
*Higher tritium concentrations in bulk salt tests allow for testing **tritium detection probes** developed by Idaho National Lab.*

*The moniker? LIBRA **One** → **TBR > 1**.*

LIBRA ONE - 700 L
in final design

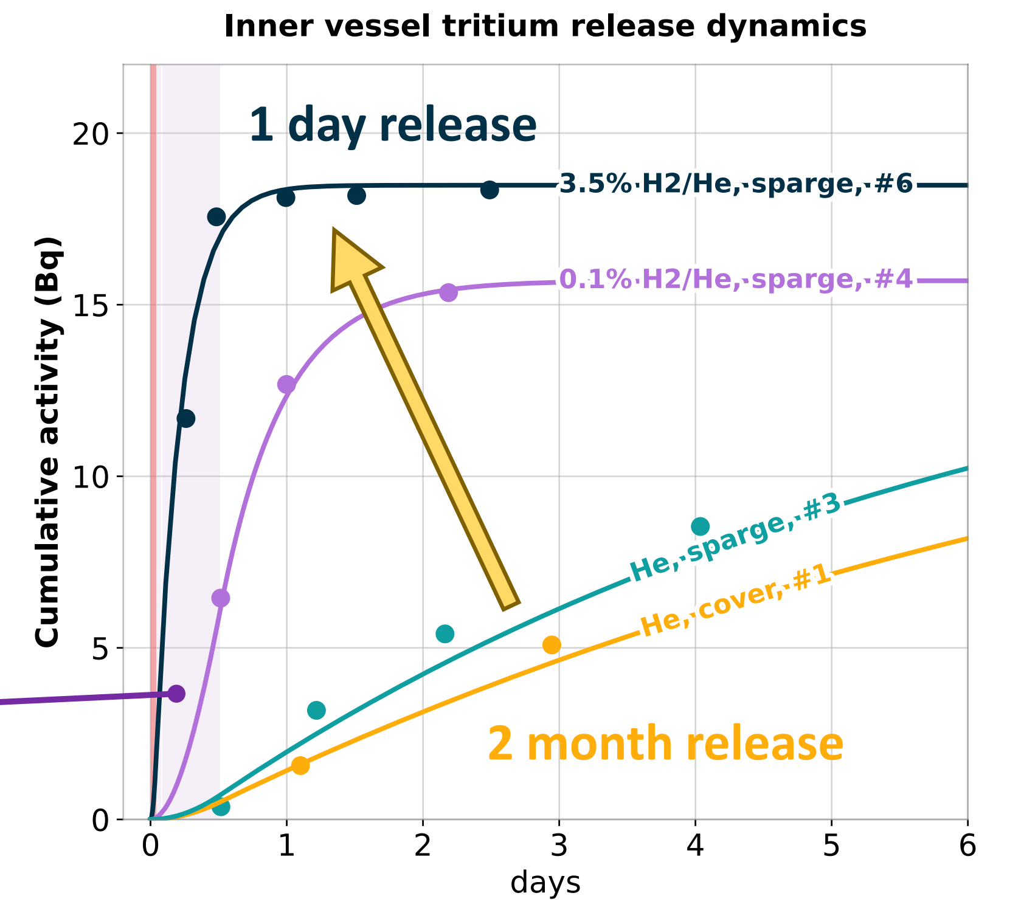
Our small-scale experiments are already yielding big results:



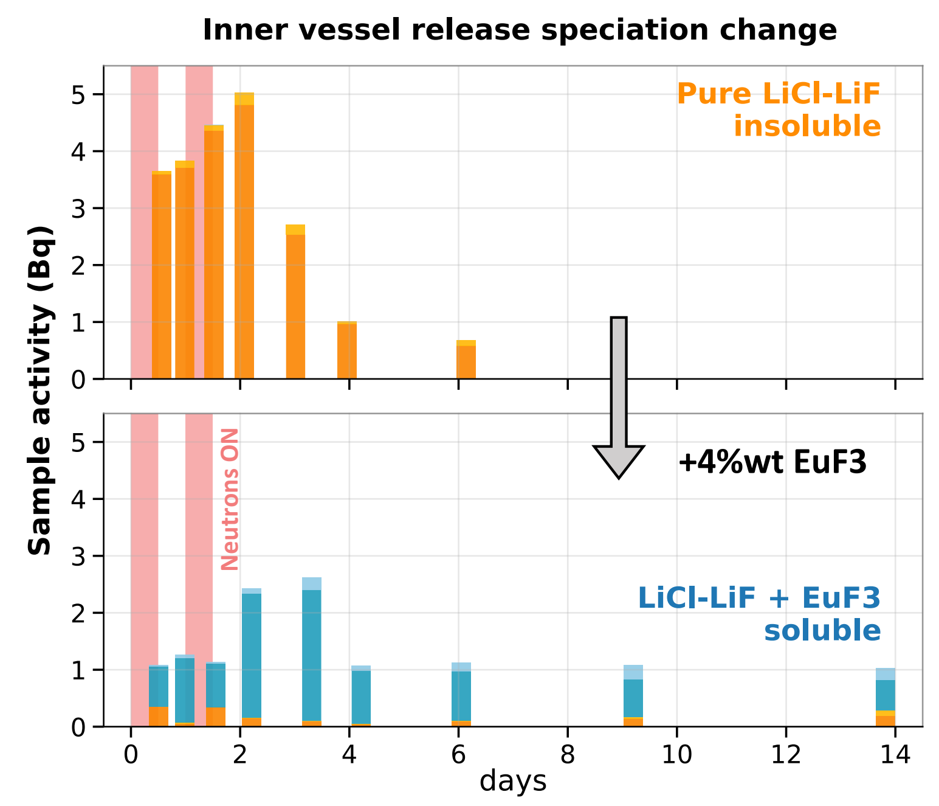


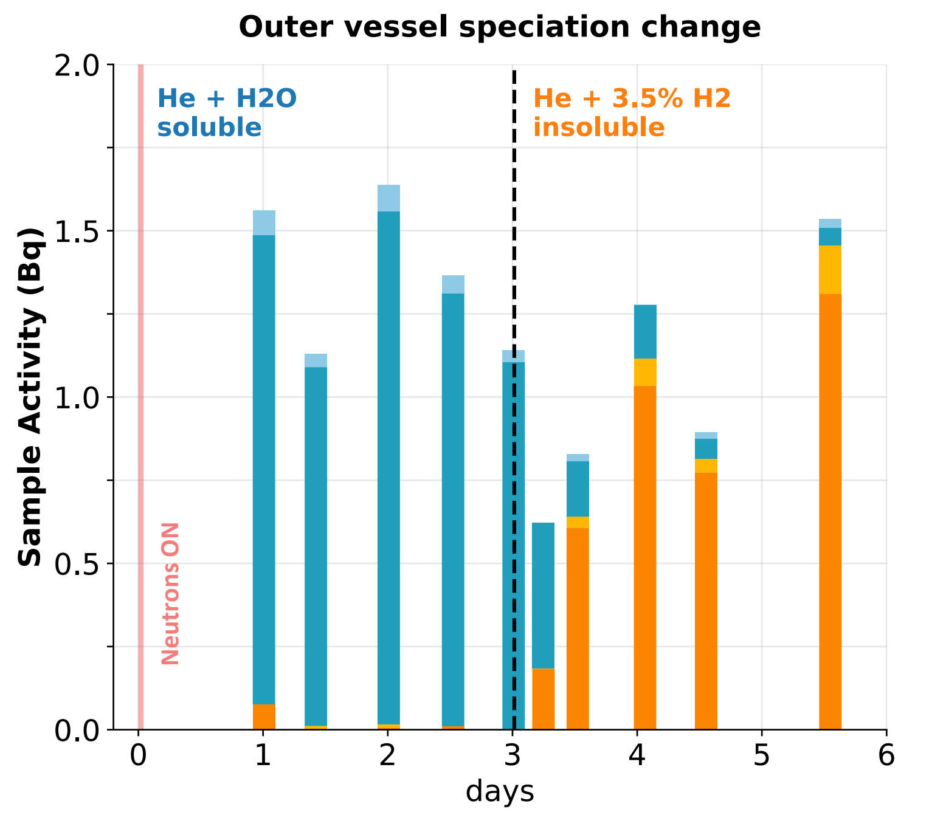
*They taught us that we can **speed up tritium release** by sparging hydrogen gas through the salt, reducing experimental time from ~2 months to ~1 day:*

*A new **Starfire nGen 6e9 neutrons/s** generator reduced the irradiation time from 12h to one hour compared to an older, 8e7 neutrons/s source.*



*We also studied the factors driving **tritium speciation**, as well as the **dynamics of permeation**:*





Recent publication on BABY: *bit.ly/100mL-BABY*

LIBRA project overview: *bit.ly/LIBRA-overview*

We are committed to open science principles and reproducibility.
*Raw data and analysis available at: **github.com/LIBRA-project***

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